Lattices of Elementary Substructures

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Discretely Ordered Semirings

$\mathbb{N}$ is a discretely ordered semiring:

- $\mathbb{N} \models \forall x \exists y \forall z (x < y \wedge (x < z \rightarrow y = z \vee y < z))$
- $\mathbb{N} \models \forall a \forall b \forall c (a \times (b + c) = (a \times b) + (a \times c))$
- ...

We can express all of this in the first order language of ordered (semi)rings: $0, 1, +, \times, <$.

- Formulas are built using variables, $0, 1, +, \times, <$ and $=$.
- Inductively: conjunctions, disjuctions, negations, and quantifications of formulas are formulas.
- Formulas can have free variables: e.g. $\phi(x) = x > 1 \wedge [\forall a \forall b (a \times b = x \implies (a = x \vee b = x))]$ expresses "x is a prime number."
- For the above formula $\phi(x)$, $\mathbb{N} \models \phi(2)$ and $\mathbb{N} \models \neg\phi(6)$.

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Elementary Extensions

Suppose $\mathcal{M} = (M, 0, 1, +, \times, <)$ and $\mathcal{N} = (N, 0, 1, +, \times, <)$ are discretely ordered semirings.

Definition

If $M \subseteq N$ and $\mathcal{M}$ and $\mathcal{N}$ agree on $+$, $\times$, and the $<$ relation for elements of M, then $\mathcal{M}$ is a substructure of $\mathcal{N}$ and $\mathcal{N}$ is an extension of $\mathcal{M}$. We write $\mathcal{M} \subseteq \mathcal{N}$.

It is not hard to see that if $\mathcal{M} \subseteq \mathcal{N}$, then $\mathcal{M}$ and $\mathcal{N}$ agree on the truth of any quantifier-free statements involving parameters from M (ie: $\mathcal{M} \models a + b = c$ if and only if $\mathcal{N} \models a + b = c$ for all $a, b, c \in M$).

If, for every formula $\phi(x_0, \ldots, x_{n-1})$ and all $a_0, \ldots, a_{n-1} \in M$, $\mathcal{M} \models \phi(\bar{a})$ if and only if $\mathcal{N} \models \phi(\bar{a})$, the extension is elementary, written $\mathcal{M} \prec \mathcal{N}$.

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Compactness

A statement is a formula without free variables. A theory is a consistent collection of statements. A model of a theory T is a structure $\mathcal{M} = (M, \ldots)$ such that all the statements in T are true in $\mathcal{M}$.

Theorem (Compactness Theorem)

Let T be a theory. T has a model if and only if every finite subset of T has a model.

Types

A type is a consistent collection of formulas in the same free variables. If $p(x)$ is a type in one free variable, and $\mathcal{M}$ is a model, then $a \in M$ realizes $p(x)$ if for each formula $\phi(x) \in p(x)$, $\mathcal{M} \models \phi(a)$.

If $\mathcal{M}$ is a model and $p(x)$ is a consistent type not realized in $\mathcal{M}$, then by the compactness theorem $\mathcal{M}$ has an elementary extension $\mathcal{N}$ with some $a \in N$ realizing $p(x)$.

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Peano Arithmetic

A model of Peano Arithmetic (PA) is a tuple $\mathcal{M} = (M, 0, 1, +, \times, <)$ such that:

1. $(M, +, \times, <)$ is a discretely ordered semiring, and
2. For each formula $\phi(x, y)$, the induction principle holds:
   $\mathcal{M} \models \forall b(\phi(0, b) \wedge \forall x(\phi(x, b) \rightarrow \phi(x+1, b))) \rightarrow \forall x \phi(x, b)$

$\mathbb{N}$ is the intended model of PA and is referred to as the standard model. Non-standard models can be constructed using ultrapowers or the compactness theorem.

Skolem Functions

Let $\phi(x, \bar{y})$ be a formula. If f is a function such that for any model $\mathcal{M}$, $\mathcal{M} \models \forall \bar{b}[\exists x \phi(x, \bar{b}) \implies \phi(f(\bar{b}), \bar{b})]$, then f is called a Skolem function.

Given a formula $\phi(x, \bar{y})$, define $\psi(x, \bar{y}) := \phi(x, \bar{y}) \wedge \forall z < x(\neg\phi(z, \bar{y}))$. $\psi(x, \bar{y})$ defines a (partial) Skolem function.

That is: PA proves that every formula has a definable Skolem function.

Skolem Closures

Given a model $\mathcal{M} \models \mathrm{PA}$ and a subset $A \subseteq M$, the Skolem closure of A, denoted $\mathrm{Scl}(A)$, is the closure of A under all Skolem functions. $\mathrm{Scl}(A) \preccurlyeq \mathcal{M}$ for any A, and is the smallest elementary substructure of $\mathcal{M}$ containing A.

Similarly, given $\mathcal{M} \prec N$ and $a \in N$, then $\mathcal{M}(a)$ is $\mathrm{Scl}(M \cup \{a\})$. This can also be thought of as the model generated by a over M. Every element in $\mathcal{M}(a)$ is of the form $f(a)$ for some $\mathcal{M}$-definable function f.

In this case: $\mathcal{M} \preccurlyeq \mathcal{M}(a) \preccurlyeq \mathcal{N}$.

Substructure and Interstructure Lattices

Definition

Given a model $\mathcal{M} \models \mathsf{PA}$, the set $\{\mathcal{K} \models \mathsf{PA} \mid \mathcal{K} \preceq \mathcal{M}\}$ is called the substructure lattice of $\mathcal{M}$ and is denoted $\mathrm{Lt}(\mathcal{M})$. If $\mathcal{M} \prec \mathcal{N}$, then the set $\{\mathcal{K} \mid \mathcal{M} \preceq \mathcal{K} \preceq \mathcal{N}\}$ is called the interstructure lattice (between $\mathcal{M}$ and $\mathcal{N}$) and is denoted $\mathrm{Lt}(\mathcal{N}/\mathcal{M})$.

Example

The simplest example is $\mathbb{N}$. $\mathrm{Lt}(\mathbb{N}) = \mathbf{1}$, the single element lattice.

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Lattices

Definition

Let L be a lattice.

- We say $a \in L$ is compact if whenever $X \subseteq L$ and $a \leq \bigvee X$, there is a finite $Y \subseteq X$ such that $a \leq \bigvee Y$.
- L is algebraic if it is complete (for all $X \subseteq L, \bigvee X$ and $\bigwedge X$ exist), and each element of L is the supremum of a set of compact elements.
- Let κ be an infinite cardinal. L is κ-algebraic if it is algebraic and, whenever $x \in L$ is compact, $|\{a \in L \mid a \leq x \text{ and } a \text{ is compact }\}| < \kappa$.

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Substructure and Interstructure Lattices II

Let $\mathcal{M} \models \mathrm{PA}$ and $\mathcal{M} \prec \mathcal{N}$. Then:

1. The compact elements of $\mathrm{Lt}(\mathcal{M})$ are the finitely generated submodels $\mathrm{Scl}(a)$ for $a \in M$.
2. Similarly, the compact elements of $\mathrm{Lt}(\mathcal{N}/\mathcal{M})$ are $\mathcal{M}(a)$ for $a \in N$.

Theorem

$\mathrm{Lt}(\mathcal{M})$ *is* $\aleph_1$*-algebraic.*

Theorem

If $\mathcal{M} \prec \mathcal{N}$ *and* $|M| = \kappa$, *then* $\mathrm{Lt}(\mathcal{N}/\mathcal{M})$ *is* κ^+*-algebraic.*

These are (so far) the only restrictions we currently have for a lattice L to be a substructure lattice or an interstructure lattice. In particular, no restrictions are known on any finite lattices.

History

- Gaifman (1960s, published 1976): Every model $\mathcal{M} \models \mathrm{PA}$ has a minimal elementary end extension.
- Gaifman: For every set I, every $\mathcal{M}$ has an elementary (end) extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong (\mathcal{P}(I), \subseteq)$.
- Mills (1979): Let D be a distributive lattice. Then D is $\aleph_1$-algebraic if and only if every $\mathcal{M} \models \mathrm{PA}$ has an elementary (end) extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong D$.

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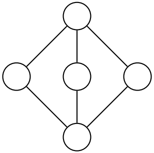

- Paris (1977) / Schmerl (1986): There is $\mathcal{M} \models \mathsf{PA}$ such that $\mathrm{Lt}(\mathcal{M}) \cong \mathbf{M}_3$.
- If $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{M}_3$, then $\mathcal{N}$ must be a cofinal extension of $\mathcal{M}$; in particular, $\mathbb{N}$ does not have an elementary extension $\mathcal{N}$ where $\mathrm{Lt}(\mathcal{M}) \cong \mathbf{M}_3$.

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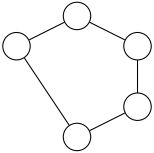

- Wilkie (1977): There is $\mathcal{M} \models \mathrm{PA}$ such that $\mathrm{Lt}(\mathcal{M}) \cong \mathbf{N}_5$. In particular, we can take an $\mathcal{M}$ such that $\mathbb{N} \prec \mathcal{M}$.
- Schmerl: there are uncountable $\mathcal{M}$ for which no $\mathcal{N} \succ_{\mathrm{end}} \mathcal{M}$ exists with $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{N}_5$.
- Open: Is it true that for every nonstandard $\mathcal{M}$, there is $\mathcal{N} \succ_{\mathrm{cof}} \mathcal{M}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{N}_5$?

Unknown

- For each n, the lattice $\mathbf{M}_n$ is the lattice with $n+2$ elements: 0, 1, and n atoms in between.
- $\mathbf{M}_1$ is the three element chain ($\mathbf{3}$) and $\mathbf{M}_2$ is the Boolean algebra on a 2-element set $\mathbf{B}_2$: both are distributive (and hence: every $\mathcal{M} \models \mathrm{PA}$ has an elementary extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{3}, \mathbf{B}_2$).
- $\mathbf{M}_n$ for $n \geq 3$ are non-distributive.

Problem (Embarrassing)

Is $\mathbf{M}_{16}$ a substructure lattice?

$n = 16$ is the smallest case for which this is unknown.

The Lattice Problem

Question

For which lattices L is it the case that there is $\mathcal{M} \models \mathrm{PA}$ such that $\mathrm{Lt}(\mathcal{M}) \cong L$?

Question

Given a model $\mathcal{M} \models \mathrm{PA}$, for which lattices L is it the case that there is $\mathcal{N}$ such that $\mathcal{M} \prec \mathcal{N}$ and $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong L$?

Theorems are often of the form: "Every (countable)/(nonstandard) $\mathcal{M}$ has an elementary (end/cofinal) extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong L$".

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Let A be any set. Then Eq(A) is the set of equivalence relations on that set, and it forms a lattice under inclusion. $\mathbf{0_A}$ is the discrete relation, and $\mathbf{1_A}$ is the trivial relation.

Definition

Let L be a finite lattice. $\alpha : L \to \mathrm{Eq}(A)$ is called a representation of L on A if:

1. α is an injection,
2. $\alpha(0) = \mathbf{1_A}$ (trivial) and $\alpha(1) = \mathbf{0_A}$ (discrete), and
3. For all $a, b \in L, \alpha(a \vee b) = \alpha(a) \wedge \alpha(b)$.

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Example: **2** (Gaifman)

Let $\mathcal{M} \models \mathsf{PA}$ be countable, $\alpha : \mathbf{2} \to \mathrm{Eq}(M)$ be the (unique) representation.

Given $\Theta \in \mathrm{Eq}(M)$ definable, we find an unbounded, definable $B \subseteq M$ such that either:

- $\Theta \cap B^2$ is trivial ($\alpha(0) \cap B^2$), or,
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Continued

Enumerate the definable equivalence relations, apply the above statement to obtain a sequence $M = A_0 \supseteq A_1 \supseteq A_2 \supseteq \ldots$. This sequence generates a type

$$p(x) = \{\phi(x) \mid \exists n < \omega\, \mathcal{M} \models \forall x \in A_n\, \phi(x)\}.$$

If c realizes $p(x)$, then $\mathcal{M} \prec_{\text{end}} \mathcal{M}(c)$ and $\text{Lt}(\mathcal{M}(c)/\mathcal{M}) \cong \mathbf{2}$.

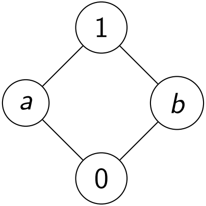
Why?

- Suppose $a \in \mathcal{M}(c)$, and f is such that $\mathcal{M}(c) \models f(c) = a$.
- Let Θ be the equivalence relation induced by f in M.
- For some n, f is either 1 to 1 or constant on A_n (hence this is in $p(x)$).
- If f is constant, then it's constantly in M, so $a \in M$, and therefore $\mathcal{M}(a) = \mathcal{M}$.
- If f is 1 to 1, then it has an inverse: $g(f(c)) = c$. So $c \in \mathcal{M}(a)$; conclude $\mathcal{M}(a) = \mathcal{M}(c)$.

The Lattice $\mathbf{B}_2$

Theorem

Let $\mathcal{M} \models \mathrm{PA}$ be countable (standard or non-standard). Then $\mathcal{M}$ has an elementary end extension $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{B}_2$.



Representation of $\mathbf{B}_2$

Let $A = \{(x, y) \mid x \leq y\}$. Define a representation $\alpha : \mathbf{B}_2 \to \mathrm{Eq}(A)$ by letting $\langle (x_0, y_0), (x_1, y_1) \rangle \in \alpha(a)$ iff $x_0 = x_1$, and $\langle (x_0, y_0), (x_1, y_1) \rangle \in \alpha(b)$ iff $y_0 = y_1$.

We need a combinatorial lemma, similar to the one for the construction of a minimal extension.

Lemma (Canonical Ramsey's Theorem (Erdős-Rado 1950))

Let $f : [\mathbb{N}]^2 \to \mathbb{N}$. There is an infinite $X \subseteq \mathbb{N}$ such that one of the following holds:

- $f \restriction [X]^2$ *is constant,*
- $f \restriction [X]^2$ *is one to one,*
- *for all* $x < y, x' < y' \in X$, $f(x, y) = f(x', y')$ *iff* $x = x'$, *or*
- *for all* $x < y, x' < y' \in X$, $f(x, y) = f(x', y')$ *iff* $y = y'$.

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Representation of $\mathbf{B}_2$

Enumerating all definable equivalence relations, we obtain a sequence of definable sets $A = A_0 \supseteq A_1 \supseteq \ldots$ and a type $p(x) = \{\phi(x) \mid \exists n < \omega\, \mathcal{M} \models \forall x \in A_n\, \phi(x)\}$

For $r \in \mathbf{B}_2$, let $t_r(x)$ be the Skolem function mapping x to the least y such that $\langle x, y \rangle \in \alpha(r)$.

Let c realize $p(x)$ and $\mathcal{N} = \mathcal{M}(c)$. Then $\mathcal{M} \prec_{\text{end}} \mathcal{N}$ and $\text{Lt}(\mathcal{N}/\mathcal{M}) \cong \mathbf{B}_2$, with the other submodels $\mathcal{M}_a$ and $\mathcal{M}_b$ being generated by $t_a(c)$ and $t_b(c)$, respectively.

The Lattice Problem

So how do we attack the lattice problem in general?

- For each finite lattice L, try to find a specific representation?
- Is there a proof that all finite lattices have n-CPP representations for every $n \in \omega$? (not yet)
- Is there a proof that all finite lattices in some infinite class have n-CPP representations for every $n \in \omega$? (Yes!)

Congruence Lattices

Let A be a set, I an index set, and for each $i \in I$, $f_i : A^n \to A$ (for some $n \in \omega$). Then $(A, \langle f_i : i \in I \rangle)$ is called an algebra.

If $\mathcal{A} = (A, \langle f_i : i \in I \rangle)$ is an algebra, then $\theta \in \mathrm{Eq}(A)$ is a congruence if it commutes with all f_i.

$\mathrm{Cg}(\mathcal{A})$ is the set of all congruences on $\mathcal{A}$; sublattice of $\mathrm{Eq}(A)$; such lattices are referred to as congruence lattices.

Congruence representations

If L is a lattice, then L^d is the lattice with the ordering reversed.

Definition

Let L be a finite lattice and $\alpha : L \to \mathrm{Eq}(A)$ be a representation. Then α is a congruence representation if there is an algebra $\mathcal{A}$ such that $\alpha \cong \mathrm{Cg}(\mathcal{A})^d$. (Such a representation is a finite congruence representation if $\mathcal{A}$ can be taken to be finite.)

Theorem (Schmerl 1993)

Suppose L is a finite lattice with a finite congruence representation. Then for every countable, nonstandard $\mathcal{M}$, there is $\mathcal{N}$ such that $\mathrm{Lt}(\mathcal{N}/\mathcal{M}) \cong L$.

Proof involves nontrivial combinatorics: Prömel-Voigt (canonical Hales-Jewett)! To illustrate its importance, we consider the example of $\mathbf{M}_3$.

The lattice $\mathbf{M}_3$

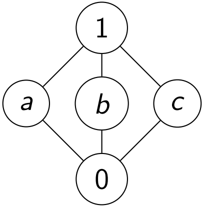

Figure: The lattice $\mathbf{M}_3$.

- $\mathbf{M}_3$ is isomorphic to the lattice of partitions of a three-element set $\{0, 1, 2\}$.
- Call this isomorphism $\alpha : \mathbf{M}_3 \to \mathrm{Eq}(3)$.
- This is a congruence representation: the algebra with no operations (or just the identity).

Powers

Let 3^m refer to the set of m-tuples of elements of $\{0, 1, 2\}$. Define $\alpha^m : \mathbf{M}_3 \to 3^m$ by $(s, t) \in \alpha^m(r)$ iff $(s_i, t_i) \in \alpha(r)$ for each $i < m$. Then:

Lemma

For every m, there is n such that whenever $\Theta \in \mathrm{Eq}(3^n)$, there is "m-dimensional" $A \subseteq 3^n$ and $r \in \mathbf{M}_3$ such that whenever $\bar{x}, \bar{y} \in A$, $(\bar{x}, \bar{y}) \in \Theta$ iff $(x_i, y_i) \in \alpha(r)$ for each $i < n$.

- m-dimensional? "There are m pairs (x_i, y_i) to check."
- Ex: $\{(a, b, 1, a, 0) : a, b \in 3\}$ is a 2-dimensional subset of 3^5.
- **Notation**: the above tells us $\alpha^n \to \alpha^m$: ie, each equivalence relation can look like $\alpha^n(r)$ for some r on an m-dimensional subset.
- How do we prove this? Previously known result in combinatorics: Prömel-Voigt / canonical Hales-Jewett.

Theorem (Hales-Jewett)

For every finite set A and every m, there is n such that every finite coloring A^n contains a monochromatic m-dimensional combinatorial subset.

There is a "canonical" version of this (akin to the "canonical" version of Ramsey's Theorem).

Theorem (Prömel-Voigt)

Let A be a finite set. For every m, there is n such that whenever $\Theta \in \mathrm{Eq}(A^n)$, there is an m-dimensional $X \subseteq A^n$ and $\theta \in \mathrm{Eq}(A)$ such that whenever $\bar{x}, \bar{y} \in X$, $(\bar{x}, \bar{y}) \in \Theta$ iff $(x_i, y_i) \in \theta$ for each $i < n$.

Look closely: for $A = \{0, 1, 2\}$, this is exactly what the Lemma on the previous slide said.

Technique

Start with $\alpha_0 = \alpha^2$. Then for each n, we can find chains $\alpha_n \to \alpha_{n-1} \to \ldots \to \alpha_0$. Overspill: $\alpha_c : \mathbf{M}_3 \to \mathrm{Eq}(3^d)$ for some nonstandard c, d. The rest is a standard technique:

- Enumerate Skolem functions $t_0, t_1, \ldots$
- Start with $A_0 = 3^d$
- Given A_i, t_i, find $r \in \mathbf{M}_3$ and "large" $A_{i+1} \subseteq A_i$ such that whenever $\bar{x}, \bar{y} \in A$, $t_i(\bar{x}) = t_i(\bar{y})$ iff $(x_j, y_j) \in \alpha(r)$ for all $j < c$.
- Now the sets $A_0 \supseteq A_1 \supseteq \ldots$ generate a type $p(x) = \{\phi(x) : \phi(M) \supseteq A_i \text{ for some } i \in \omega\}$.
- If a realizes $p(x)$, then $\mathrm{Lt}(\mathcal{M}(a)/\mathcal{M}) \cong \mathbf{M}_3$.

Known and unknown

- Known: every algebraic lattice is isomorphic to a congruence algebra.
- Open ("finite lattice representation problem"): is every finite lattice isomorphic to $\mathrm{Cg}(\mathcal{A})$ for some finite $\mathcal{A}$?
- Every finite lattice that is currently known to be a substructure lattice has a finite congruence representation.
- Unknown: is there a finite substructure lattice that does not have a finite congruence representation?

Thank you!

Some references:

- Abdul-Quader and Kossak, "The Lattice Problem for Models of PA." Bulletin of Symbolic Logic (2025).
- Kossak and Schmerl, *The Structure of Models of Peano Arithmetic* (Chapter 4)
- Schmerl, "Substructure lattices of models of Peano arithmetic." Logic Colloquium 84, pp. 225-243 (1986).
- Schmerl, "Finite substructure lattices of models of Peano arithmetic." Proceedings of the American Mathematical Society (1993).